Year 10 Mathematics

Number 2

4.1 Percentages

Percentages

STARTER · 5 MIN

Date: _____ **Title:** Percentages

LEARNING OBJECTIVES

- Increase and decrease amounts using a multiplier
- Find a percentage increase or decrease
- Find the original amount (reverse percentages)

STARTER *Answer in your book. No calculators.*

1. Work out 10% of 60
2. Work out 25% of 80
3. Write 0.35 as a percentage
4. Write 15% as a decimal

Starter — Answers

STARTER

Tick or correct in green pen

1. 10% of $60 =$ **6**
2. $80 \div 4 =$ **20**
3. $0.35 \times 100 =$ **35%**
4. $15 \div 100 =$ **0.15**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can increase and decrease amounts using a multiplier	☐
SILVER	I can find a percentage increase or decrease	☐
GOLD	I can find the original amount (reverse percentages)	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Percentages

INTRODUCTION · 3 MIN

Use a multiplier: one calculation does the whole job.

Increase

Add the percentage to 100%.

Increase by 20%: × 1.20

Decrease

Take the percentage away from 100%.

Decrease by 15%: × 0.85

Percentage change

Change ÷ original × 100

$$\frac{\text{change}}{\text{original}} \times 100$$

Reverse percentage

Divide the new amount by the multiplier.

Original = new ÷ multiplier

Examples 1 and 2 — Using Multipliers

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint new amount = original × multiplier

Example 1

Increase £80 by 15%

Example 2

Decrease 250 kg by 12%

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Multiplier = 1.15

80 × 1.15 **= £92**

Tip: 100% + 15% = 115% = 1.15

Example 2

Multiplier = 0.88

250 × 0.88 **= 220 kg**

Tip: 100% − 12% = 88% = 0.88

Task 1

PRACTICE · 9 MIN

BRONZE *Use a multiplier for each question*

Hint new amount = original × multiplier

1. Increase £60 by 20%
2. Decrease 150 by 30%
3. Increase 45 kg by 8%
4. Decrease £320 by 5%
5. A coat costs £75. The price rises by 12%. Find the new price.

★ Write down the multiplier for an increase of 3.5%

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $60 \times 1.2 =$ **£72**
2. $150 \times 0.7 =$ **105**
3. $45 \times 1.08 =$ **48.6 kg**
4. $320 \times 0.95 =$ **£304**
5. $75 \times 1.12 =$ **£84**

★ $1 + 0.035 =$ **1.035**

Examples 3 and 4 — Percentage Change

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint $\%\ \text{change} = \dfrac{\text{change}}{\text{original}} \times 100$

Example 3

The price of a book rises from £40 to £46.

Find the percentage increase.

Example 4

A car is bought for £12 000 and sold for £9000.

Find the percentage loss.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$\text{Change} = 46 - 40$$
$$= 6$$
$$\text{Increase} = \frac{6}{40} \times 100$$

Increase **= 15%**

Tip: always divide by the original amount.

Example 4

$$\text{Change} = 12\,000 - 9000$$
$$= 3000$$
$$\text{Loss} = \frac{3000}{12000} \times 100$$

Loss **= 25%**

Tip: the original amount is the price paid.

Task 2

PRACTICE · 9 MIN

SILVER *Find the percentage increase or decrease*

Hint $\%\ \text{change} = \frac{\text{change}}{\text{original}} \times 100$

1. 50 to 65

2. 80 to 68

3. £250 to £280

4. 1.6 kg to 1.2 kg

5. A shop buys a phone for £150 and sells it for £195. Find the percentage profit.

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. Change $= 15$, $\frac{15}{50} \times 100 =$ **30% increase**

2. Change $= 12$, $\frac{12}{80} \times 100 =$ **15% decrease**

3. Change $= 30$, $\frac{30}{250} \times 100 =$ **12% increase**

4. Change $= 0.4$, $\frac{0.4}{1.6} \times 100 =$ **25% decrease**

5. Profit $= 45$, $\frac{45}{150} \times 100 =$ **30% profit**

Examples 5 and 6 — Reverse Percentages

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint original = $\frac{\text{new amount}}{\text{multiplier}}$

Example 5

(3 marks)

After a 20% increase, the price of a bag is £72.

Find the original price.

Example 6

(3 marks)

In a sale, prices are reduced by 15%.

A TV costs £340 in the sale.

Find the price before the sale.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Multiplier = 1.2

Original = 72 ÷ 1.2

Original **= £60**

Tip: divide, do not take 20% off £72.

Example 6

Multiplier = 0.85

Original = 340 ÷ 0.85

Original **= £400**

Tip: 85% of the original is £340.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint original = $\frac{\text{new amount}}{\text{multiplier}}$

1. After a 25% increase, a price is £150. Find the original price. (3 marks)

2. After a 10% decrease, a mass is 54 kg. Find the original mass. (3 marks)

3. After a 4% pay rise, a salary is £31 200. Find the salary before the rise. (3 marks)

4. A jacket costs £63 after 30% off. Find the price before the reduction. (3 marks)

5. £2000 is invested at 3% compound interest per year. Find the value after 2 years. (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $150 \div 1.25 =$ **£120**

2. $54 \div 0.9 =$ **60 kg**

3. $31\,200 \div 1.04 =$ **£30 000**

4. $63 \div 0.7 =$ **£90**

5. $2000 \times 1.03^2 =$ **£2121.80**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Increase £40 by 35%

SILVER Find the percentage increase from 25 to 30

GOLD After a 40% reduction, a game costs £36. Find the original price.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $40 \times 1.35 =$ **£54**

SILVER Change $= 5$, $\frac{5}{25} \times 100 =$ **20%**

GOLD $36 \div 0.6 =$ **£60**

Exam Practice — Multipliers

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint new amount = original × multiplier

Question 1

(3 marks)

Write down the decimal multiplier for

(a) an increase of 15% (1)

.....

(b) a decrease of 8% (1)

.....

(c) an increase of 2.5% (1)

.....

Question 2

(2 marks)

A laptop normally costs $640.
In a sale, the price is reduced by 15%.
Work out the sale price of the laptop.



..... $

Exam Practice — Percentage Change

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint $\%\ \text{change} = \dfrac{\text{change}}{\text{original}} \times 100$

Question 3

(3 marks)

The population of a town increased from 2400 to 2760.

Work out the percentage increase.

..... %

Question 4

(3 marks)

The table shows what Sam paid for a car and what he sold it for.

Work out Sam's percentage loss.

Bought for	Sold for
£12 000	£9300

..... %

Exam Practice — Reverse Percentages

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint original = $\frac{\text{new amount}}{\text{multiplier}}$

Question 5

(3 marks)

In a sale, all prices are reduced by 35%.

The sale price of a coat is £52.

Work out the original price of the coat.



..... £

Question 6

(3 marks)

The value of a car decreases by 12% each year.

After one year it is worth $15 840.

Work out the value of the car when new.

..... $

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 1.15 B1

(b) 0.92 B1

(c) 1.025 B1

Tip: increase: 1 + r, decrease: 1 − r.

Question 2

640 × 0.85 M1

= $544 A1

Tip: 100% − 15% = 85% → × 0.85.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

2760 − 2400 = 360	M1
360 ÷ 2400 × 100	M1
= 15%	A1

Tip: change ÷ ORIGINAL × 100.

Question 4

12 000 − 9300 = 2700	M1
2700 ÷ 12 000 × 100	M1
= 22.5%	A1

Tip: divide by the price he PAID.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

100% − 35% = 65%	M1
52 ÷ 0.65	M1
= £80	A1

Tip: divide by the multiplier, never × 1.35.

Question 6

0.88 × value = 15 840	M1
15 840 ÷ 0.88	M1
= $18 000	A1

Tip: check: 18 000 × 0.88 = 15 840 ✓